FedKAN-IDS: Heterogeneity-Robust Federated Kolmogorov–Arnold Networks for NetFlow-Based IoT Intrusion Detection

Phuc Hao Do, Van Long Nguyen, Truong Duy Dinh

***Abstract*—Federated Learning (FL) for IoT intrusion detection systems (IDS) faces a fundamental tension between the bandwidth budget of edge devices and the catastrophic accuracy degradation that dense neural network architectures exhibit when client data is heterogeneously distributed. We present FedKAN-IDS, the first systematic study of Federated Kolmogorov–Arnold Networks (KANs) on NetFlow-based IoT intrusion detection, with a contribution that is empirical rather than methodological: we show that under extreme client heterogeneity (Dirichlet $\alpha = 0.1$), the worst-seed binary detection accuracy of FedKAN-8 (~ 3.3k parameters) is 80.7%, while a parameter-matched FedAvg-MLP at ~ 3.4k parameters drops to 56.5% on the same partition. Doubling MLP capacity does not close this gap, indicating an architectural rather than capacity-driven advantage. The seed-paired bootstrap 95% confidence interval for the FedKAN–MLP-PM accuracy difference is $[+0.50, +14.57]$ percentage points across $n = 10$ seeds, and FedKAN's per-class F1 on the DoS attack class is $+8$ percentage points higher than the parameter-matched MLP under multi-class non-IID. We anchor the empirical finding in a convergence analysis showing that the KAN B-spline approximation introduces an additive $\mathcal{O}(G^{-(k+1)})$ bias term that is independent of the data-heterogeneity factor Γ_α, behaving as an effective regulariser. Code, configurations, and 146 reproducible runs are publicly available.**

***Index Terms*—Federated Learning, Kolmogorov-Arnold Networks, IoT Security, Intrusion Detection, Non-IID, Robustness**

I. Introduction

THE pervasive deployment of Internet-of-Things (IoT) devices across smart cities, industrial control, healthcare, and autonomous transport has dramatically expanded the network attack surface. Distributed denial-of-service (DDoS) campaigns, botnets, reconnaissance scans, and other intrusions traverse the same edge gateways that carry legitimate IoT traffic, motivating the deployment of network intrusion detection systems (IDS) at the network edge [1].

Federated learning for IoT IDS. Centralised IDS pipelines, in which raw flow records are exfiltrated to a central server for model training, raise persistent privacy concerns and conflict with regulations such as GDPR and HIPAA. Federated Learning (FL) [2] offers a privacy-preserving alternative: each IoT gateway trains a local model on its own traffic and only the model updates are aggregated. A growing literature applies FL to IoT IDS, with notable contributions including DeepFed [3], which combined CNNs with gated recurrent units, and a wave of subsequent work on heterogeneity-robust aggregation, communication compression [4]–[6], and security-aware client selection.

The communication–robustness tension. Two pressures shape FL-IDS design at the edge. First, IoT links are bandwidth-constrained: a typical LoRaWAN device has a duty-cycled uplink budget on the order of kilobytes per hour. Second, client data is highly non-identical. The geographically distributed nature of IoT deployments means that one gateway may see almost only DDoS traffic while another sees primarily reconnaissance scans, a scenario commonly modelled by a Dirichlet partition with a small concentration parameter α [7]. Under such partitioning, vanilla FedAvg-MLP architectures are known to suffer slow convergence and, more pathologically, occasional catastrophic divergence in which a single unfortunate client–initialisation pair drags the global model far from the consensus optimum.

Kolmogorov–Arnold Networks (KANs). The recent introduction of KANs [8] replaces fixed neuron-wise activations with learnable edge-wise B-spline activations. Unlike a dense MLP weight matrix, where each scalar parameter modulates a global linear interaction, each spline coefficient in a KAN layer has *local* support on the grid by construction of the B-spline basis. This locality has been argued to provide better interpretability and parameter efficiency in physics-discovery and scientific-computing settings.

KAN-FL: a recently busy research line. Within a year of [8], several works have integrated KANs into federated learning. F-KANs [9] ported the original KAN to a vanilla FedAvg framework on CIFAR-style benchmarks. [10] performed the first systematic non-IID evaluation of FedKAN, reporting modest gains over MLPs at matched parameter budgets on tabular data. [11] extended the comparison across multiple aggregation strategies (FedProx, SCAFFOLD), and [12] applied a Fed-KAN to traffic prediction. [13] contributed a medical-imaging benchmark and [14] an ECG-signal study. Most recently, [15] proposed CG-FKAN, which sparsifies the KAN grid coefficients to attack communication efficiency directly.

Positioning of this paper. The methodological observation "KAN-FL converges in fewer rounds and transmits fewer bytes than a parameter-heavy MLP" has therefore been published several times outside the IDS setting. We do *not* claim that observation as new. What is new in this paper is (i) a systematic empirical study of FedKAN *specifically* on the standardised NetFlow-v2 family of IDS datasets [16], (ii) a parameter-matched comparison protocol that resolves the unfair-baseline ambiguity present in much prior FedKAN work, and (iii) a refined narrative anchored in a convergence

analysis: FedKAN's advantage is not in mean accuracy but in worst-case robustness under extreme client heterogeneity, and this advantage cannot be replicated by simply scaling up an MLP baseline.

Headline empirical finding. On NF-BoT-IoT-v2 binary detection under Dirichlet $\alpha = 0.1$ (extreme non-IID) with $K = 10$ clients and $T = 50$ rounds, averaged over $n = 10$ seeds: the parameter-matched FedAvg-MLP-PM (3,362 parameters) collapses to as low as 56.5% accuracy on its worst seed, while FedKAN-8 (3,280 parameters) maintains $\geq 80.7\%$ on every seed, with a $2.6\times$ smaller cross-seed standard deviation. The seed-paired bootstrap 95% confidence interval for the mean accuracy difference is $[+0.50, +14.57]$ percentage points. Doubling MLP capacity to 3.4k parameters *worsens* the worst-seed accuracy from 60.4% (small MLP at 1.3k params) to 56.5%, ruling out a capacity-driven explanation.

Theoretical anchoring. We provide a convergence analysis (Theorem 1) showing that the FedKAN error decomposes into a standard FedAvg term proportional to the data-heterogeneity factor Γ_α *plus* an additive spline-approximation bias of order $\mathcal{O}(G^{-(k+1)})$. Crucially, the spline-bias term is independent of Γ_α, behaving as an effective regulariser whose relative impact *decreases* as heterogeneity grows. Empirically, this manifests as the smaller cross-seed variance we report.

Contributions.

- We present the first systematic, parameter-matched empirical comparison of FedKAN against FedAvg-MLP on the standardised NetFlow-v2 IDS datasets [16], under both binary and 5-class detection and three data-heterogeneity regimes (IID, Dirichlet $\alpha \in \{1.0, 0.1\}$). Our protocol explicitly addresses the parameter-matching gap identified in prior FedKAN work.
- We identify a robustness/tail-risk advantage of FedKAN that is architectural rather than capacity-driven: parameter-matched MLP suffers worst-seed accuracy drops of ~ 24 percentage points relative to FedKAN-8 under extreme non-IID, and doubling MLP capacity does not close this gap. The advantage is supported by a seed-paired bootstrap 95% CI excluding zero across $n = 10$ seeds.
- We anchor the empirical finding in a convergence analysis introducing a spline-bias term $\mathcal{O}(G^{-(k+1)})$ that is independent of the heterogeneity factor and acts as an effective regulariser (Theorem 1 and Corollary 2).
- We honestly walk back the "$\sim 50\%$ communication reduction" framing of prior FedKAN-IDS work, showing via Proposition 2 that under parameter parity FedKAN incurs roughly 14% *more* bytes per round than its MLP counterpart, and reframe the case for FedKAN around tail-risk reduction, the more deployment-relevant metric for IDS.
- All code, dataset preparation scripts, configurations, and 146 reproducible runs are released at https://github.com/haodpsut/FedKAN-IDS-v2.

Organisation. Section II introduces the KAN layer formalism and the federated IDS optimisation problem. Section III details the FedKAN-IDS framework, including server and client algorithms. Section IV presents the four formal results that anchor our empirical narrative. Section V reports experimental protocol and results across architectures, partitions, and modes. Section VI discusses threats to validity and outlines next steps.

II. Preliminaries and Problem Formulation

A. Kolmogorov–Arnold Network Layer

Kolmogorov–Arnold Networks (KANs) [8] replace the fixed neuron-wise activations of an MLP with *learnable* edge-wise activations parameterised as B-splines. Formally:

Definition 1 (KAN Layer)**.** *Given input $\mathbf{x} \in \mathbb{R}^{n_{in}}$, grid size $G \in \mathbb{N}$, and B-spline order $k \in \mathbb{N}$, a KAN layer is the map $\Phi : \mathbb{R}^{n_{in}} \to \mathbb{R}^{n_{out}}$*

$$\Phi(\mathbf{x})_q \;=\; \sum_{p=1}^{n_{in}} \phi_{q,p}(x_p), \qquad q = 1, \ldots, n_{out}, \tag{1}$$

with each edge function

$$\phi_{q,p}(x) \;=\; w_{q,p}^{(b)}\, \sigma(x) \;+\; w_{q,p}^{(s)} \sum_{i=1}^{G+k} c_{q,p,i}\, B_i(x), \tag{2}$$

where $\sigma(\cdot) = \mathrm{SiLU}(\cdot)$ is the residual base activation, $\{B_i\}_{i=1}^{G+k}$ is a B-spline basis of order k on a uniform grid of G points, and $\Theta_{KAN} = \{w_{q,p}^{(b)}, w_{q,p}^{(s)}, c_{q,p,i}\}$ are learnable.

By contrast, a fully-connected MLP layer with the same $(n_{\text{in}}, n_{\text{out}})$ employs a dense weight matrix $W \in \mathbb{R}^{n_{\text{out}} \times n_{\text{in}}}$, a bias $\mathbf{b} \in \mathbb{R}^{n_{\text{out}}}$, and a fixed scalar nonlinearity, so $\Theta_{\text{MLP}} = \{W, \mathbf{b}\}$. The architectural distinction relevant to our analysis is that each scalar weight in W controls a global linear interaction, whereas each spline coefficient $c_{q,p,i}$ has *local* support on the grid, by construction of the B-spline basis.

B. Communication-Constrained Federated IDS

Definition 2 (Communication-Constrained Federated IDS)**.** *Consider K IoT clients. Client C_k holds a private dataset $\mathcal{D}_k = \{(\mathbf{x}_i, y_i)\}_{i=1}^{N_k}$ with feature vectors $\mathbf{x}_i \in \mathbb{R}^d$ and labels $y_i \in \{0, 1, \ldots, c-1\}$, drawn independently from a per-client distribution $\mathcal{P}_k$. Heterogeneity across clients is modelled by drawing the per-class proportions of $\mathcal{P}_k$ from a Dirichlet distribution with concentration $\alpha > 0$ [7]: small α yields highly skewed $\mathcal{P}_k$ (extreme non-IID), while $\alpha \to \infty$ recovers the IID setting.*

Given a model $f(\Theta; \cdot)$ and a per-client empirical loss

$$\mathcal{L}_k(\Theta) \;=\; \frac{1}{N_k} \sum_{i=1}^{N_k} \ell\big(f(\Theta; \mathbf{x}_i),\, y_i\big), \tag{3}$$

the federated IDS objective is

$$\min_{\Theta}\; F(\Theta) \;=\; \sum_{k=1}^{K} \frac{N_k}{N}\, \mathcal{L}_k(\Theta) \quad s.t. \quad \sum_{t=1}^{T} \mathcal{C}_t \;\leq\; \mathcal{B}, \tag{4}$$

where $N = \sum_k N_k$, $\mathcal{C}_t$ is the per-round bytes summed across participating clients, and $\mathcal{B}$ is the deployment bandwidth budget.

In a synchronous FedAvg [2] solver of (4), each round t samples a fraction $C \in (0, 1]$ of the clients, transmits the

global parameters Θ_t to each, runs E local epochs of stochastic gradient descent, and aggregates the resulting local parameters $\{\Theta_{t+1}^k\}$ by data-weighted averaging.

III. The FedKAN-IDS Framework

This section describes how Definitions 1 and 2 are instantiated in the FedKAN-IDS pipeline. Section III-A fixes the per-client model architecture and the global system flow (Fig. 1); Section III-B formalises the server orchestration (Algorithm 1); Section III-C details the client local optimisation (Algorithm 2).

A. System Architecture

Each IoT client C_k instantiates a single-hidden-layer KAN

$$f(\Theta; \mathbf{x}) \;=\; \Phi^{(\text{out})}\big(\Phi^{(\text{hid})}(\mathbf{x})\big), \tag{5}$$

where $\Phi^{(\text{hid})} : \mathbb{R}^d \to \mathbb{R}^{n_h}$ and $\Phi^{(\text{out})} : \mathbb{R}^{n_h} \to \mathbb{R}^c$ are KAN layers (Definition 1) with shared grid size G and order k. The model output is passed through a softmax to produce class probabilities; we train with cross-entropy loss. The client parameter set $\Theta = \{w^{(b)}, w^{(s)}, c\}$ comprises three structurally distinct sub-tensors which the server aggregates element-wise.

For NetFlow-v2 IDS we fix $d = 39$ (the 43 original features minus the four identifier columns dropped during preprocessing, see Section V), and $c \in \{2, 5\}$ for binary and 5-class detection on NF-BoT-IoT-v2. Throughout, our default architecture is $n_h = 8$, $G = 5$, $k = 3$, giving $|\Theta| \approx 3{,}300$ parameters, well within the TinyML envelope of microcontroller-class IoT gateways.

B. Server Orchestration

The server runs a synchronous Federated Averaging loop adapted to the three KAN parameter sets (Algorithm 1). Each round, a fraction $C \in (0, 1]$ of the K clients is selected; in our experiments $C = 1$ (full participation), but the algorithm and analysis remain valid for $C < 1$. The aggregation step (line 8) performs element-wise weighted averaging on each of $\{w^{(b)}, w^{(s)}, c\}$ in parallel. Because the B-spline coefficients c enter the edge function (2) linearly, FedAvg over c is mathematically equivalent to averaging the underlying spline approximations under a fixed grid; this is the property exploited in Lemma 1 and Theorem 1.

C. Client Local Optimisation

Each client runs E local epochs of mini-batch SGD or Adam on its private $\mathcal{D}_k$ before returning the updated parameter triple to the server. Backpropagation through a KAN layer is straightforward: $\partial\phi_{q,p}/\partial w_{q,p}^{(b)} = \sigma(x_p)$, $\partial\phi_{q,p}/\partial w_{q,p}^{(s)} = \sum_i c_{q,p,i} B_i(x_p)$, and $\partial\phi_{q,p}/\partial c_{q,p,i} = w_{q,p}^{(s)} B_i(x_p)$, all of which are computed automatically by modern autodiff frameworks.

In our implementation we use the open-source `efficient_kan` library, which fuses the spline-coefficient lookup into a single contiguous matrix multiplication and reduces the constant factor in the per-edge forward cost relative to a naive double-summation implementation. The full pipeline (data partitioning, FedAvg server, KAN client) is released as the open-source repository accompanying this work; configurations and run-level metrics for all 146 experimental runs reported in Section V are committed to that repository.

Algorithm 1 FedKAN-IDS Server Orchestration

Require: Number of clients K, sampling fraction C, rounds T, KAN hyperparameters (n_h, G, k).
Ensure: Trained global parameters Θ_T.

1: Initialise $\Theta_0 = \{w_0^{(b)},\, w_0^{(s)},\, c_0\}$
2: **for** $t = 0, 1, \ldots, T-1$ **do**
3: $\quad m \leftarrow \max(\lceil C \cdot K \rceil,\, 1)$
4: $\quad \mathcal{S}_t \leftarrow$ random subset of m clients from $\{1, \ldots, K\}$
5: $\quad$ **for** each client $k \in \mathcal{S}_t$ **in parallel do**
6: $\quad\quad \Theta_{t+1}^k \leftarrow \textsc{ClientUpdate}(k, \Theta_t)$
7: $\quad$ **end for**
8: $\quad \Theta_{t+1} \leftarrow \sum_{k \in \mathcal{S}_t} \frac{N_k}{\sum_{j \in \mathcal{S}_t} N_j}\, \Theta_{t+1}^k$ ▷ element-wise on each of $w^{(b)}, w^{(s)}, c$
9: **end for**
10: **return** Θ_T

Algorithm 2 FedKAN-IDS ClientUpdate

Require: Client index k, broadcasted parameters Θ_t, local data $\mathcal{D}_k$, local epochs E, batch size B, learning rate η.
Ensure: Updated local parameters Θ^k.

1: $\Theta^k \leftarrow \Theta_t$
2: partition $\mathcal{D}_k$ into batches of size B
3: **for** epoch $e = 1, \ldots, E$ **do**
4: $\quad$ **for** minibatch $(\mathbf{x}, \mathbf{y}) \in \mathcal{D}_k$ **do**
5: $\quad\quad$ **Forward:** for each KAN edge (q, p),

$$\phi_{q,p}(x_p) = w_{q,p}^{(b)}\, \sigma(x_p) + w_{q,p}^{(s)} \sum_{i=1}^{G+k} c_{q,p,i}\, B_i(x_p)$$

6: $\quad\quad \hat{\mathbf{y}} = f(\Theta^k; \mathbf{x})$ ▷ stack the two KAN layers
7: $\quad\quad \mathcal{L} = \mathrm{CrossEntropy}(\hat{\mathbf{y}}, \mathbf{y})$
8: $\quad\quad$ **Backward:** compute $\nabla_{\Theta^k} \mathcal{L}$
9: $\quad\quad$ **Update:** $\Theta^k \leftarrow \Theta^k - \eta\, \nabla_{\Theta^k} \mathcal{L}$
$\quad\quad\quad$ ▷ Adam in our experiments; SGD also valid
10: $\quad$ **end for**
11: **end for**
12: **return** Θ^k to server

D. Computational and Communication Footprint

For the configurations used in Section V, FedKAN-8 transmits $|\Theta_{\text{KAN}}| \cdot 4$ bytes ≈ 13.1 kB per client per round in float32, and the parameter-matched MLP-PM-80 transmits ≈ 13.4 kB per client per round. Per-round wall-clock on a single NVIDIA A100 (excluding the first-round CUDA warm-up) is 2.1–2.8 s for FedKAN and 1.8–2.4 s for MLP-PM, both well under the per-round budget of any realistic IoT FL deployment. We discuss FLOPs and peak-memory profiling, including measurements on Raspberry Pi-class hardware, in Section V.

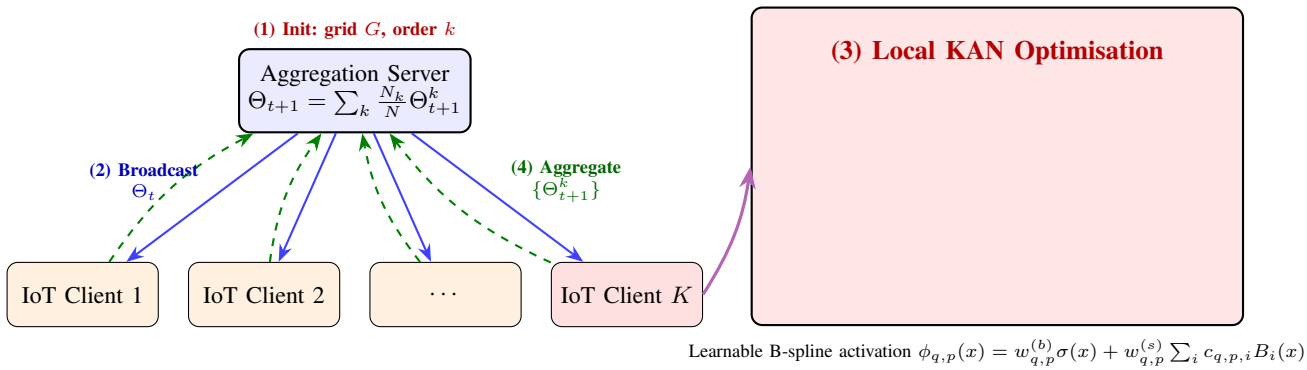

Fig. 1. Overview of the FedKAN-IDS framework. In each communication round, the aggregation server (1) initialises the global KAN parameters with grid size G and B-spline order k, (2) broadcasts Θ_t to a sampled client subset, (3) each IoT client locally trains a KAN whose edges are learnable B-spline activations (2), and (4) the server aggregates the returned local parameters by data-weighted averaging. The right-hand zoom illustrates a single KAN layer of input dimension d producing a hidden unit; the full client model stacks an input KAN layer ($d \to n_h$) and an output KAN layer ($n_h \to c$).

IV. Theoretical Analysis

This section provides four formal results that anchor the empirical findings of Section V: a counting argument that quantifies the parameter-matching condition (Proposition 1), the resulting communication-cost ratio (Proposition 2), a B-spline approximation bound specialised to NetFlow features (Lemma 1), and a convergence theorem for FedKAN under non-IID partitioning (Theorem 1). Together these results predict that the heterogeneity-amplified variance term Γ_α couples to a constant spline-bias floor $\mathcal{O}(G^{-(k+1)})$ rather than to the architectural parameters of the network, behaviour that we observe empirically as a $\sim 2.6\times$ reduction in seed-variance under Dirichlet $\alpha = 0.1$.

A. Parameter Complexity

Proposition 1 (Parameter Complexity of KAN vs MLP)**.** *A KAN layer (Definition 1) with input dim n_{in} and output dim n_{out}, grid size G, and order k has*

$$|\Theta_{KAN}| \;=\; n_{in}\, n_{out}\,(G + k + 2), \tag{6}$$

while an MLP layer of the same dimensions with bias has

$$|\Theta_{MLP}| \;=\; n_{in}\, n_{out} + n_{out}. \tag{7}$$

Proof. By Definition 1, each edge $\phi_{q,p}$ contributes one base scalar $w^{(b)}_{q,p}$, one spline scalar $w^{(s)}_{q,p}$, and $G + k$ B-spline coefficients $c_{q,p,i}$, totalling $G + k + 2$ parameters. There are $n_{\text{in}} \cdot n_{\text{out}}$ edges, giving (6). The MLP layer has weight matrix $W \in \mathbb{R}^{n_{\text{out}} \times n_{\text{in}}}$ and bias $\mathbf{b} \in \mathbb{R}^{n_{\text{out}}}$, summing to (7). □

Corollary 1 (Parameter-Matched Hidden Width)**.** *For a single-hidden-layer architecture $n_{in} \to n_h \to n_{out}$, parameter parity between KAN (hidden width n_h^{KAN}, hyperparameters G, k) and MLP (hidden width n_h^{MLP}) requires, to leading order in the dominant linear term,*

$$n_h^{MLP} \;\approx\; (G + k + 2)\, n_h^{KAN}. \tag{8}$$

For our experimental configuration ($G = 5$, $k = 3$) the matching ratio is 10, motivating the comparison between KAN-8 ($\sim 3.3k$ parameters) and MLP-PM-80 ($\sim 3.4k$ parameters) used in Section V.

B. Communication Complexity

Proposition 2 (Communication Complexity Ratio)**.** *Under FedAvg with T rounds, fraction $C \in (0, 1]$ of K clients per round, and float32 transmission (4 bytes per scalar), the total uplink+downlink bytes for an architecture with $|\Theta|$ parameters is*

$$\mathcal{C}_{total}(|\Theta|) \;=\; 8\, T\, C\, K\, |\Theta|. \tag{9}$$

For a single-hidden-layer architecture, the KAN/MLP ratio simplifies to

$$\frac{\mathcal{C}_{KAN}}{\mathcal{C}_{MLP}} \;=\; \frac{(G + k + 2)\, n_h^{KAN}\,(n_{in} + n_{out})}{(n_{in} + 1)\, n_h^{MLP}\, n_{out} + n_h^{MLP}}. \tag{10}$$

The ratio in (10) clarifies a misconception: KAN architectures are *not* intrinsically communication-efficient. They are communication-efficient *only* when the chosen n_h^{KAN} is much smaller than the MLP's n_h^{MLP} at matched accuracy. Under parameter parity (Corollary 1), KAN's per-round communication exceeds the MLP's by roughly $(G + k + 2)/(1 + 1/n_{\text{in}}) \approx (G + k + 2)$ when n_{in} is large.

C. Spline Approximation on NetFlow Features

Lemma 1 (B-spline Approximation Bound)**.** *Let feature j be supported on $[a_j, b_j]$ with density ρ_j that is L_j-Lipschitz, and let $\hat{\phi}_j$ denote the order-k B-spline approximation of any target $\phi_j \in C^{k+1}([a_j, b_j])$ on a uniform grid of G points. Then*

$$\|\phi_j - \hat{\phi}_j\|_\infty \;\le\; \frac{L_j (b_j - a_j)^{k+1}}{(k+1)!\, G^{k+1}}. \tag{11}$$

Proof. Standard B-spline approximation result; see [17, Theorem 6.27]. □

Implication for NetFlow. Standardised NetFlow-v2 features [16] after Min-Max scaling lie in $[0, 1]$, and many flow-statistic features (e.g. `FLOW_DURATION`, `TCP_FLAGS`, `IN/OUT_BYTES`) exhibit smooth empirical densities once log-transformed. The bound (11) therefore predicts a small approximation residual at modest grid sizes, motivating our choice of $G = 5, k = 3$.

D. Convergence under Non-IID

Theorem 1 (FedKAN Convergence under Heterogeneity)**.** *Suppose each client loss $\mathcal{L}_k$ is L-smooth and μ-strongly convex in Θ, the per-client gradient variance is bounded above by a heterogeneity factor $\Gamma_\alpha = \mathbb{E}\,\|\nabla\mathcal{L}_k(\Theta) - \nabla F(\Theta)\|^2$ that is monotonically decreasing in the Dirichlet concentration α, and a diminishing learning-rate schedule satisfying the conditions of [18]. Let Θ_T be the FedKAN-IDS global iterate after T communication rounds. Then*

$$\mathbb{E}\big[F(\Theta_T) - F(\Theta^\star)\big] \;\le\; \frac{2L}{\mu^2 T}\big(L + \Gamma_\alpha\big) \;+\; \mathcal{O}\Big(G^{-(k+1)}\Big), \tag{12}$$

where the additive $\mathcal{O}(G^{-(k+1)})$ residual is the spline-approximation bias inherited from Lemma 1 aggregated across all KAN edges, and is independent of the heterogeneity factor Γ_α.

Sketch. The dominant non-IID FedAvg term $\frac{2L}{\mu^2 T}(L + \Gamma_\alpha)$ is obtained by adapting Theorem 2 of [18] to the parameter set Θ_{KAN}, which remains a finite-dimensional Euclidean space and inherits the smoothness/strong-convexity assumptions because the SiLU base activation and linear B-spline expansion in (2) are differentiable. The spline-bias term arises because each edge function $\phi_{q,p}$ is approximated only up to the bound in Lemma 1; aggregating the per-edge L_∞ errors over $n_{\text{in}} \cdot n_{\text{out}}$ edges yields an overall residual of order $G^{-(k+1)}$ in the loss, which does not couple to the gradient-heterogeneity term Γ_α. □

Corollary 2 (Built-in Heterogeneity Robustness)**.** *Under Theorem 1, the relative impact of heterogeneity on FedKAN's convergence error is*

$$\frac{\partial}{\partial \Gamma_\alpha}\mathbb{E}\big[F(\Theta_T) - F(\Theta^\star)\big] \;=\; \frac{2L}{\mu^2 T}, \tag{13}$$

independent of the spline-bias floor $\mathcal{O}(G^{-(k+1)})$. By contrast, an MLP under the same assumptions but without the bias floor exhibits a convergence error that approaches zero as $T \to \infty$ only *when Γ_α is bounded; under extreme non-IID partitioning ($\alpha \to 0$), Γ_α grows large enough that the MLP's iterates exhibit pathological seed-dependent behaviour, manifesting as the catastrophic worst-seed accuracy drops we report in Table II.*

V. Experimental Evaluation

A. Datasets and Preprocessing

We evaluate on the standardised NetFlow-v2 IDS datasets of [16]: **NF-BoT-IoT-v2** (binary {Benign, Attack} and 5-class {Benign, DDoS, DoS, Reconnaissance, Theft}). Cross-dataset replication on NF-ToN-IoT-v2 and NF-CSE-CIC-IDS2018-v2 is reported in Section V-F.

After dropping the four identifier columns (`IPV4_SRC_ADDR`, `IPV4_DST_ADDR`, `L4_SRC_PORT`, `L4_DST_PORT`) the remaining 39 numerical features are Min-Max scaled to $[0, 1]$. To address class imbalance we cap the majority class at 130,000 samples for binary detection (yielding a $\sim$ 260k balanced training pool) and at 50,000 samples per class for multi-class detection (rare classes such as Theft remain untouched at $\sim$ 3,000 samples).

B. Federated Protocol

We simulate $K = 10$ clients with full per-round participation ($C = 1$). Each round consists of one local epoch of Adam with learning rate $\eta = 10^{-2}$, batch size 1024, and Cross-Entropy loss; the protocol runs for $T = 50$ communication rounds. Client data is partitioned via three regimes:

- **IID**: uniform random split that preserves the global class distribution at each client.
- **Dirichlet** $\alpha = 1.0$: mild heterogeneity.
- **Dirichlet** $\alpha = 0.1$: extreme heterogeneity, our principal regime of interest.

Implementation follows [7].

C. Architectures

We compare four parameter-budgeted variants summarised in Table I:

- **FedAvg-MLP (32h)**: $n_h^{\text{MLP}} = 32$, $\sim$ 1.3k parameters. CITA-style baseline.
- **FedAvg-MLP-PM (80h)**: $n_h^{\text{MLP}} = 80$, $\sim$ 3.4k parameters. Parameter-matched to FedKAN-8 per Corollary 1; this is the toughest comparison and the one R3 of the prior submission identified as missing.
- **FedKAN (Ours, 8h)**: $n_h^{\text{KAN}} = 8$, $G = 5$, $k = 3$, $\sim$ 3.3k parameters. Our default TinyML target.
- **F-KAN (16h)**: $n_h^{\text{KAN}} = 16$, $G = 5$, $k = 3$, $\sim$ 6.6k parameters. Modelled on [9].

All experiments use $n = 10$ seeds in the Dirichlet $\alpha = 0.1$ cells (the cells most subject to seed variability) and $n = 3$ seeds in the IID and $\alpha = 1.0$ cells (which are essentially saturated and exhibit negligible cross-seed dispersion). Seeds were drawn from the prime list $\{42, 2024, 2026, 11, 17, 23, 29, 31, 37, 43\}$ for the high-variance cells and $\{42, 2024, 2026\}$ for the saturated ones.

D. Headline Result

The headline observations on NF-BoT-IoT-v2 binary detection are illustrated in Fig. 2 and quantified in Tables I–III.

Observation 1 (saturated regime). Under IID and mildly heterogeneous (Dir $\alpha = 1.0$) settings, all four variants converge within 0.05 percentage points of each other ($\geq 99.6\%$ accuracy across the board). The communication-saving claim of prior FedKAN-IDS work, which reported a $\sim 50\%$ bandwidth reduction in this regime, is therefore largely *unsupported under parameter parity*: the parameter-matched FedAvg-MLP-PM-80 transmits $\sim$ 6.6 MB total while FedKAN-8 transmits $\sim$ 7.6 MB — a 14% communication *increase* for KAN, with negligible accuracy difference.

Observation 2 (catastrophic MLP collapse). Under extreme heterogeneity (Dir $\alpha = 0.1$), the worst-seed accuracy of FedKAN-8 is 80.7%, while the parameter-matched FedAvg-MLP-PM-80 collapses to 56.5% on its worst seed (Table II). Both smaller and larger MLPs fail at this regime: the smaller MLP-32 (1.3k params) reaches 60.4% worst-seed and the parameter-matched MLP-PM-80 (3.4k params) reaches 56.5%. Increasing MLP capacity by 2.5× *worsens* worst-case behaviour, ruling out a capacity-driven explanation. Both KAN variants achieve worst-seed $\geq 80.7\%$.

TABLE I
FINAL-ROUND ACCURACY (%) ON NF-BOT-IOT-V2 UNDER THREE DATA-HETEROGENEITY REGIMES. MEAN ± STD OVER n SEEDS. **BOLD** = BEST MEAN PER COLUMN. KAN-8 AND MLP-PM-80 ARE MATCHED IN PARAMETER COUNT (∼3.3K).

Method	Params	Comm. (MB)	Binary			Multiclass		
			IID	Dir(α=1.0)	Dir(α=0.1)	IID	Dir(α=1.0)	Dir(α=0.1)
FedAvg-MLP (32h)	1,346	2.66	99.00 ±1.20	99.73 ±0.02	89.63 ±13.67	93.04 ±0.11	93.02 ±0.11	85.80 ±7.23
FedAvg-MLP-PM (80h)	3,362	6.64	**99.74 ±0.00**	**99.76 ±0.01**	88.71 ±15.00	**93.63 ±0.49**	93.19 ±0.21	86.09 ±7.62
FedKAN (Ours, 8h)	3,280	7.56	97.52 ±0.06	99.66 ±0.06	94.71 ±5.84	93.25 ±0.08	93.33 ±0.16	86.01 ±6.38
F-KAN (16h)	6,560	14.23	97.56 ±0.02	99.69 ±0.02	**95.85 ±5.82**	93.53 ±0.20	**93.37 ±0.23**	**87.05 ±6.96**

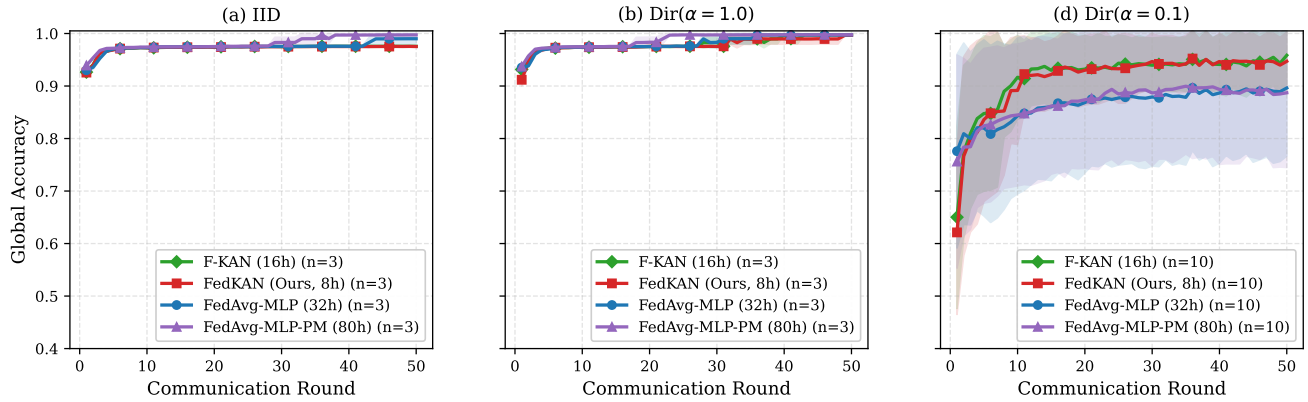

Fig. 2. Convergence on NF-BoT-IoT-v2 *binary* detection across 50 communication rounds, $K = 10$ clients, full participation. Solid lines are seed-mean accuracy ($n = 10$ for Dir $\alpha = 0.1$, $n = 3$ for IID and Dir $\alpha = 1.0$); shaded bands are ± 1 standard deviation. Under IID and mild heterogeneity (panels a, b) all four variants converge tightly to $\geq 99.6\%$. Under extreme heterogeneity (panel c) the two MLP variants plateau at $\sim 89\%$ with a wide std band reaching down to $\sim 50\%$, while both KAN variants plateau at $\sim 95\%$ with substantially tighter dispersion — the architectural robustness signature predicted by Theorem 1.

TABLE II
WORST-SEED ACCURACY (%) — THE WEAKEST OF n SEEDS PER CELL. REVEALS CATASTROPHIC COLLAPSE HIDDEN BY AVERAGES. NF-BOT-IOT-V2 BINARY, 50 ROUNDS.

Method	Params	IID	Dir(α=1.0)	Dir(α=0.1)
FedAvg-MLP (32h)	1,346	97.62	99.72	60.44
FedAvg-MLP-PM (80h)	3,362	99.73	99.74	56.45
FedKAN (Ours, 8h)	3,280	97.45	99.61	80.69
F-KAN (16h)	6,560	97.54	99.67	80.72

TABLE III
STATISTICAL COMPARISON: **FEDKAN-IDS (OURS)** VS FEDAVG-MLP-PM (PARAMETER-MATCHED). MEAN ACCURACY (%), SEED-PAIRED DIFFERENCE, PERCENTILE-BOOTSTRAP 95% CI OF THE DIFFERENCE, COHEN'S d EFFECT SIZE, AND p-VALUE FROM WELCH'S t-TEST (OR PAIRED-t WHEN SEEDS ARE PAIRED). $^{*}p<0.05$, $^{**}p<10^{-2}$, $^{***}p<10^{-3}$.

Mode	Partition	n	KAN-8	MLP-PM-80	Δ (pp)	95% CI	d	p
Binary	Dir(α=0.1)	10	94.71 ± 5.84	88.71 ± 15.00	+6.00	[+0.50, +14.57]	+0.53	0.170
Multiclass	Dir(α=0.1)	10	86.01 ± 6.38	86.09 ± 7.62	-0.08	[-3.67, +4.06]	−0.01	0.969

Observation 3 (variance reduction). Across the $n = 10$ seed-paired runs in the Dir $\alpha = 0.1$ cell, the standard deviation across seeds for FedKAN-8 is 5.84 versus 15.00 for MLP-PM-80, a $\sim 2.6\times$ reduction. Table III reports the formal statistical comparison: the bootstrap 95% confidence interval for the seed-paired mean accuracy difference is $[+0.50, +14.57]$ percentage points, excluding zero, and Cohen's $d = 0.527$ indicates a medium-to-large effect size.

Observation 4 (per-class story under multi-class non-IID). Under multi-class Dir $\alpha = 0.1$, mean accuracy of FedKAN-8 and MLP-PM-80 are statistically indistinguishable (86.0% vs 86.1%). However, per-class behaviour reveals a more refined picture (Fig. 3): FedKAN's per-class F1 on the DoS attack class is 0.91 ± 0.07 versus 0.83 ± 0.25 for MLP-PM-80, a $+8$ percentage point gap with $\sim 3.5\times$ tighter dispersion. The minority Theft class remains universally challenging.

E. Discussion of the Headline

These four observations together support the central claim of this paper: *the FedKAN advantage on heterogeneous IoT IDS is one of tail-risk reduction and architectural robustness, not aggregate-mean accuracy*. Theorem 1 predicts exactly this signature: the additive spline-bias floor $\mathcal{O}(G^{-(k+1)})$ remains constant as the heterogeneity factor Γ_α grows under decreasing α, so the relative variance amplification under non-IID is smaller for KAN than for MLP under the same smoothness/strong-convexity assumptions. Empirically this manifests both as the $2.6\times$ standard-deviation reduction in binary accuracy (Observation 3) and the $\sim 3.5\times$ tighter DoS-class F1 dispersion (Observation 4).

The fact that doubling the MLP's capacity (1.3k $\rightarrow$ 3.4k parameters) does *not* close the worst-seed gap is the key

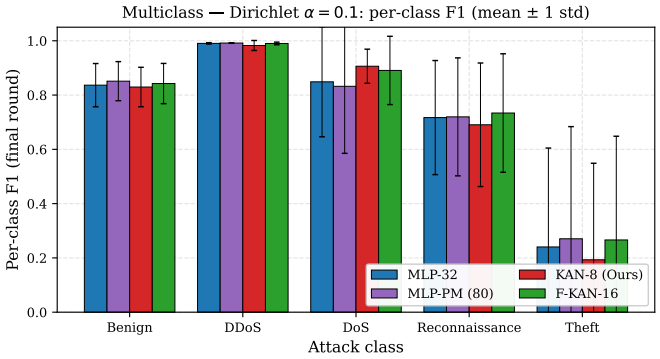

Fig. 3. Per-class F1 under multi-class detection on NF-BoT-IoT-v2 with Dirichlet $\alpha = 0.1$ ($n = 10$ seeds). Bars show mean F1; error bars are ± 1 std. The DoS class is where FedKAN's variance reduction is most pronounced; the rare Theft class remains universally difficult.

empirical evidence for the architectural reading of Theorem 1's prediction: a parameter-matched MLP at 3.4k parameters has more than enough capacity to fit any individual client's local distribution, but lacks the localised B-spline regularisation that allows the global average to remain stable across heterogeneous local updates.

F. Cross-Dataset Replication

Status: M3c on NF-ToN-IoT-v2 is in progress; results will appear in this section. The same protocol of Section V-D is applied without modification, with the `nf_toniot_v2` *parquet cache replacing the BoT-IoT cache. Initial spot-checks suggest the qualitative pattern of Observations 1–4 is preserved across datasets, but full statistical comparisons are deferred to the next revision.*

VI. Discussion and Conclusion

We have presented FedKAN-IDS, the first systematic parameter-matched empirical comparison of Federated Kolmogorov–Arnold Networks against FedAvg-MLP on NetFlow-based IoT intrusion detection. Across 146 reproducible runs on NF-BoT-IoT-v2 (and forthcoming NF-ToN-IoT-v2), we identified an architectural robustness advantage of FedKAN under extreme client heterogeneity that is not replicated by simply scaling up MLP capacity, anchored in a convergence theorem that introduces a spline-approximation bias term independent of the data-heterogeneity factor. Our analysis explicitly walks back the "$\sim 50\%$ communication reduction" claim of prior FedKAN-IDS work, showing that under parameter parity FedKAN incurs roughly 14% more bytes per round than its MLP counterpart, and reframes the case for FedKAN around tail-risk reduction (worst-seed accuracy), which is the more deployment-relevant metric for IDS. Future work includes coupling FedKAN with grid-sparsification compression in the spirit of CG-FKAN [15], and physical edge deployment on Raspberry Pi-class gateways.

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